

Operator's Manual
The Coherent Infinity™
Nd:YAG Laser System



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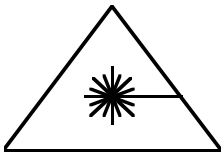
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Operation of any Coherent laser with any of its interlocks defeated is always at the operator's own risk.

Operating Information

Operating the laser without understanding the following information may cause extensive damage to the system that is not covered by the warranty. Read the Operator's Manual for more detailed information. Keep this document with the laser at all times.



Direct eye contact with the output beam from the laser may cause permanent blindness. Laser safety glasses must be worn at all times when operating the laser.



Lasing is immediately terminated and all systems in the laser are sequentially switched off if an interlock is tripped. To restart the laser, close the interlock, clear the fault message and press "On" in the main control window.



By manually increasing the flash lamp voltage the Infinity system can deliver output energies significantly higher than its specifications. Doing this will, however, permanently damage optical components in the amplifier. Do not run the system at higher energies or powers than what it is rated for.



Modifying the parameters in the Lamp Timer window will affect laser performance. Deviation from factory settings may result in lower output power and unstable DPSS laser performance.



Operate the laser at maximum specified output energy for minimum intensity noise and optimum mode quality.



It is important that the depolarized energy, displayed in the Monitor window, is kept at a minimum. If it is above 1.5 %, of the IR pulse energy, see Procedure 4 in Chapter Four.



All optical components must be cleaned in their mounts. Removing and reinserting any of the optical components in the laser will severely affect the mode quality and may require a complete realignment to be performed. This can only be done by a Coherent Field Service Engineer. Do not attempt to optimize the output power by adjusting the alignment of the laser.



Set the Lamp max voltage in the main Control window to be only 5–10 V higher than what is needed to produce the required output power. This prevents the energy servo from applying excessive voltage to the flash lamp in case the beam is partially blocked.



The resistivity of the internal cooling water, displayed in the Monitor window, should be greater than 2 MΩ/cm. If necessary adjust the bypass valve in the power supply. Replace the deionization cartridge when a resistivity higher than 2 MΩ/cm can not be obtained even with the bypass valve fully open.



Use only steam distilled water in the internal water cooling system. Do not use deionized tap water.



If the internal water cooling system is drained, remove the deionization cartridge and seal it so that it does not dry out. If the cartridge is allowed to dry it will lose its capacity to maintain a high enough water resistivity. Do not insert a new cartridge until you are ready to fill the system with distilled water.



Make sure that the beam is never retro-reflected back into the laser. Even a very weak return reflex can be amplified by residual gain in the Nd:YAG rods and destroy parts of the laser. If a reflected pulse is too intense a fault message stating “Depolarized energy too high” appears and the laser is automatically turned off. The same pulse that causes the laser to turn off itself might, however, also damage the system.



The material in the Infinity beam dump can safely absorb 42 W of average power of IR. It can thus be used for the IR power produced by any combination of pulse energy and repetition rate. It is, however, not designed to absorb any more of the harmonic radiation than what goes through the respective dichroics. It is therefore important to observe the following:

1. The beam dump can only handle the residual IR + the harmonic light that leaks through the dichroics.
 2. When installing a harmonic option, the harmonic crystal must not be left detuned for any time longer than necessary to complete the alignment procedure. Damage caused by directly dumping harmonic radiation (produced at 532 nm, 355 nm and 266 nm) into the beam dump is not covered by the warranty. Always use the SHG, THG and FHG crystals with their respective dichroics.
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Table 1. Glossary of Units

UNIT	QUANTITY
AC	Alternating Current
Amp	Amperes
°C	Degrees Centigrade
DC	Direct Current
°F	Degrees Fahrenheit
Hz	Hertz
J	Joule
lbs	Pounds
mrاد	milliradians = 10^{-3} radians
nm	nanometers = 10^{-9} m
psi	Pounds per square inch
V	Volts
W	Watts

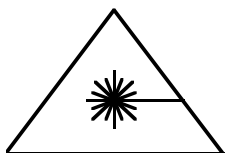
Table 2. Glossary of Abbreviations

ABBREVIATION	MEANING
AOM	Acousto-Optical Modulator
BBO	Beta Barium Borate
DPMO	Diode Pumped Master Oscillator
DPSS	Diode Pumped Solid State Laser
FHG	Fourth Harmonic Generation
IR	Infrared
LED	Light Emitting Diode
PCM	Phase Conjugate Mirror
SBS	Stimulated Brillouin Scattering
SHG	Second Harmonic Generation
SLM	Single Longitudinal Mode
THG	Third Harmonic Generation

Preface

Congratulations on your purchase of the Infinity Nd:YAG laser. We at Coherent hope that this laser will become a valuable tool for you. The Infinity laser system represents a major breakthrough in pulsed high power solid state laser technology. Its design is based on a diode pumped master oscillator, followed by a multi-pass power amplifier stage that incorporates a phase conjugate mirror, serving as a corrective element for phase distortions. As a result the Infinity offers true single longitudinal mode operation, diffraction-limited brightness, near top-hat intensity distribution and a repetition rate that is continuously variable from single-shot up to 100 Hz for 40-100 systems, 50 Hz for 25-50 systems, and 30 Hz for 15-30 systems.

As with all sophisticated equipment, proper maintenance can significantly extend the life span of components and also ensure that the Infinity laser continuously will perform at the specified level. It is therefore strongly recommended that you allocate some time to study this manual before operating the system. By studying this manual you will gain a better understanding of how the laser works. This translates into less down time, a more reliable system, and better performance. Problems will be detected earlier and the range of applications extended. If you have any questions about the laser, its operation or the content of this manual, please do not hesitate to contact Coherents Service Department at 800-367-7890 (within the US), 408-764-4557 (outside the US) or your local Coherent representative.



Use of controls, adjustments, or performance of procedures other than those specified herein, may result in hazardous radiation exposure.



Read this manual carefully before operating the laser for the first time. Special attention should be given to the material in Chapter One, Laser Safety.

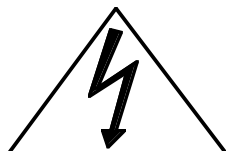
U.S. Export Control Laws Compliance

It is the policy of Coherent to comply strictly with U.S. export control laws.

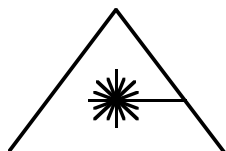
Export and re-export of lasers manufactured by Coherent are subject to U.S. Export Administration Regulations, which are administered by the Commerce Department. In addition, shipments of certain components are regulated by the State Department under the International Traffic in Arms Regulations.

The applicable restrictions vary depending on the specific product involved and its destination. In some cases, U.S. law requires that U.S. Government approval be obtained prior to resale, export or re-export of certain articles. When there is uncertainty about the obligations imposed by U.S. law, clarification should be obtained from Coherent or an appropriate U.S. Government agency.

Symbols Used In This Document



This symbol is intended to alert you to the presence of dangerous voltage within the product's enclosure that may be of sufficient magnitude to constitute a risk of electric shock and to indicate possible risk of equipment damage.



This symbol is intended to alert you to the danger of exposure to hazardous visible and invisible laser radiation.



This symbol is intended to emphasize the presence of important operating and maintenance instructions.
